

ABSTRACT

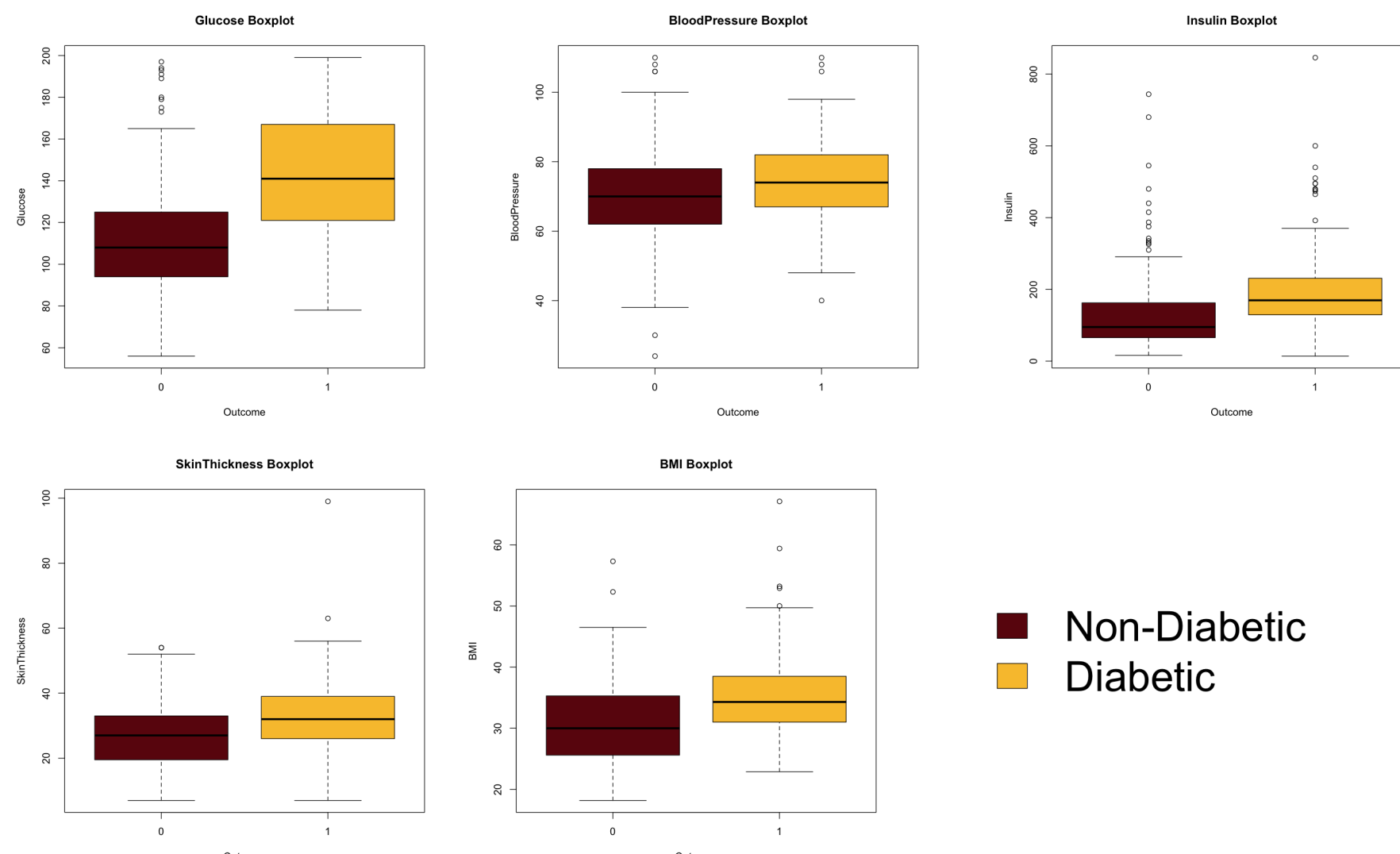
Diabetes is a chronic disease that affects millions of people worldwide, and early prediction is important for improving prevention and treatment outcomes. In this study, we explore data-driven methods to build a reliable prediction model using a real-world diabetes dataset.

Keywords: Diabetes, Machine Learning, Logistic Regression, K-Means, SVM, KNN

DATA PREPROCESSING

To improve data quality and model performance, several preprocessing techniques were applied to the Pima Indians Diabetes dataset:

- Mean Imputation



Missing values (zeros in clinical measurements) were replaced using mean imputation, producing more reasonable variable distributions.

- PCA

All predictors were standardized using Z-Score Scaling prior to PCA. The first 5 principal components were retained, explaining approximately 80% of the total variance.

- K-Means Clustering

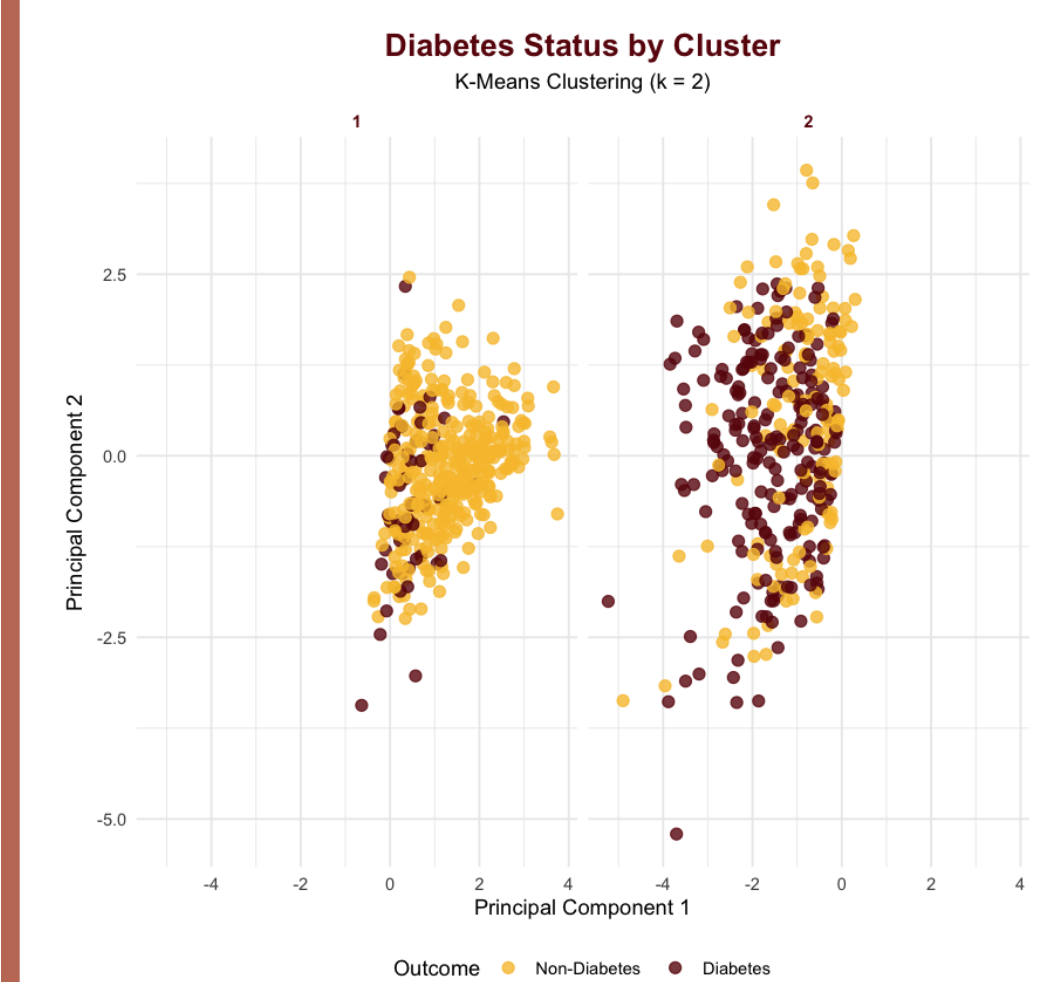
For K-means clustering, the optimal number of clusters was determined using the Elbow and Silhouette methods, resulting in $k=2$. Cluster assignments were then used for noise removal, where observations in the top 5% of Euclidean distance from cluster centroids were treated as outliers and removed.

FUTURE RESEARCH

- Explore additional preprocessing techniques.
- Compare deep learning models with traditional machine learning.
- Investigate feature selection methods.
- Evaluate performance on larger datasets.

K-MEANS CLUSTERING + LR

K-means Clustering Visualization ($k = 2$)



We selected $k = 2$ based on clustering analysis. PCA was used to reduce the original eight predictors to two principal components, providing a two-dimensional visualization of the clustering results. The plot illustrates the separation of the two clusters and the distribution of diabetes status.

Some predictors in the full model were not statistically significant at the 5% level, indicating model redundancy and motivating feature reduction.

Stepwise Logistic Regression Model

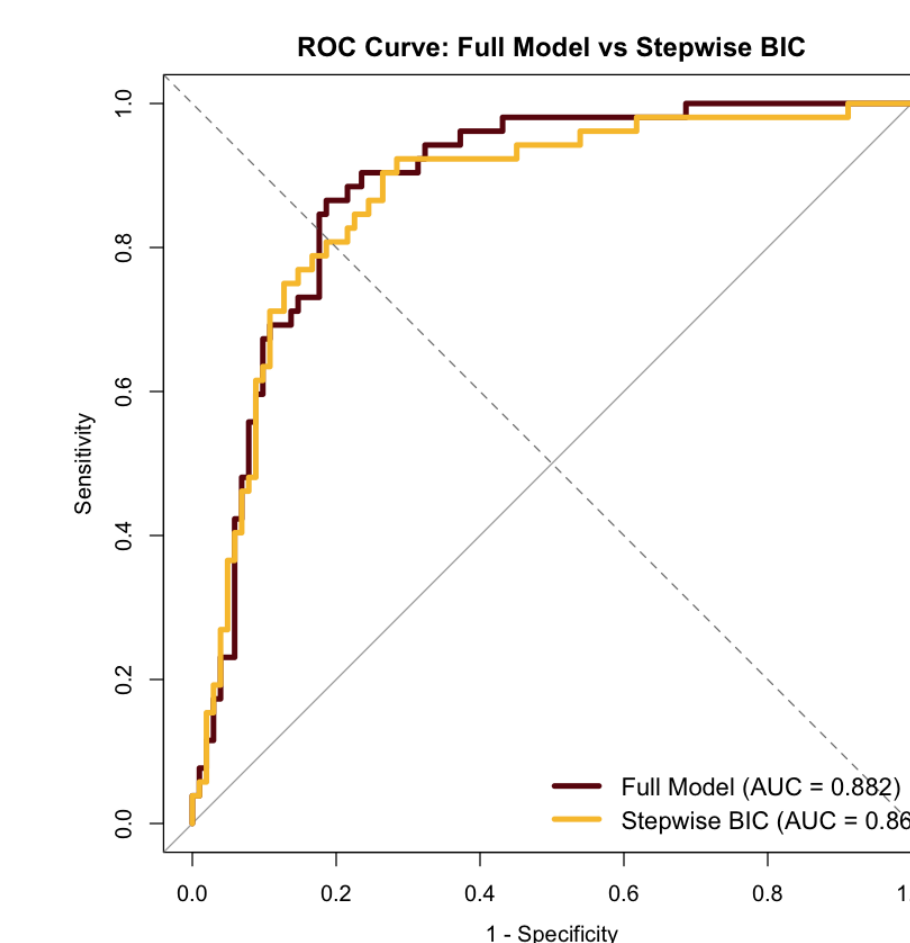
Predictor	Coefficient	p-value
Pregnancies	0.354	< 0.001
Glucose	0.954	< 0.001
Skin Thickness	0.457	< 0.001
Insulin	1.306	< 0.001

The stepwise model retains only statistically significant predictors, reducing model complexity while maintaining performance.

Metric	Overall Model	Stepwise Model
Accuracy	0.8117	0.8182
Precision	0.7255	0.74
Recall	0.7115	0.7115
F1 Score	0.7184	0.7255
AIC	507.01	509.93
BIC	546.32	531.77
AUC	0.882	0.867

Model Comparison

Both full and stepwise models achieved strong performance. Stepwise model achieves similar predictive ability with fewer variables. It is preferred due to better interpretability and parsimony.



SVM

Support Vector Machine (SVM) is a supervised classification algorithm that separates classes by finding the optimal decision boundary.

SVM was used to evaluate the effectiveness of our preprocessing methods by classifying diabetes outcomes and comparing performance using accuracy, precision, recall, F1-score, and ROC/AUC.

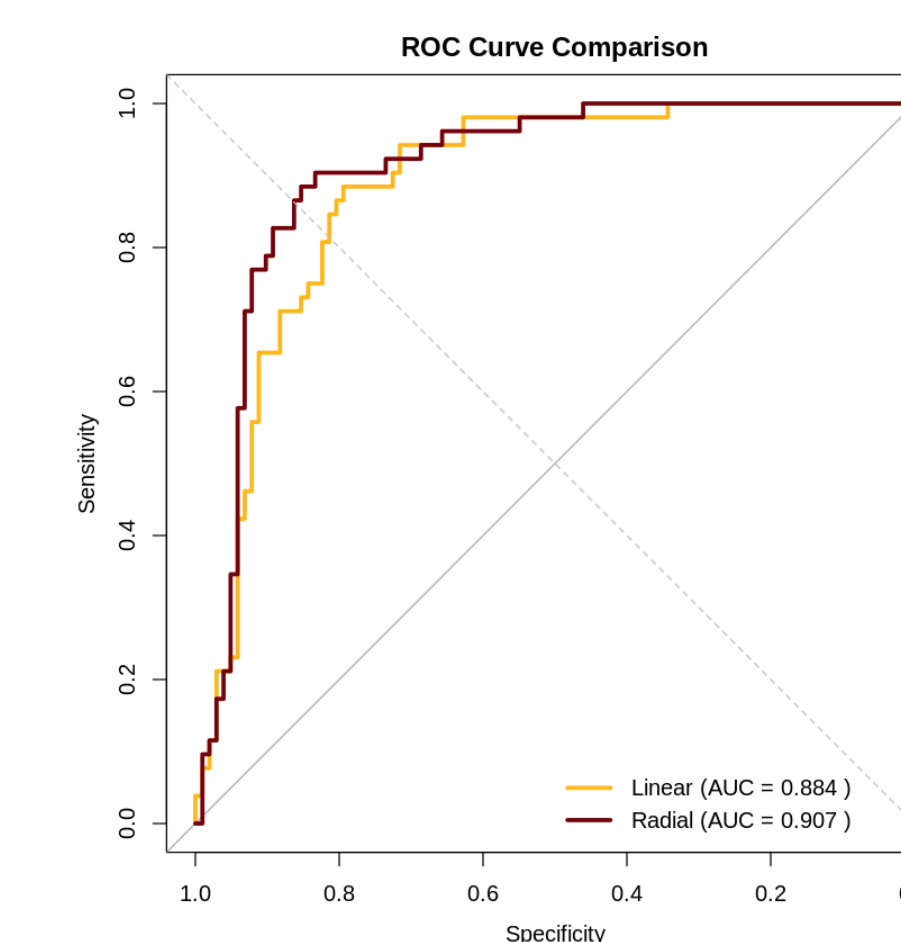
Linear and Radial (Gaussian RBF) kernels were evaluated to compare classification performance after preprocessing.

Metric	Linear	Radial
Accuracy	0.8182	0.8636
Precision	0.75	0.8163
Recall	0.6923	0.7692
F1 Score	0.72	0.792
AUC	0.884	0.907

Model Comparison

The Radial (RBF) SVM outperformed the Linear SVM across all evaluation metrics.

ROC analysis also showed improved discrimination, with the Radial SVM achieving a higher AUC (0.907) than the Linear SVM (0.884).



RESULTS

Model Comparison

Model	Acc	Prec	Recall	AIC	BIC
LR	0.7857	0.721	0.5962	560.47	600.25
PCA + LR	0.7857	0.7111	0.6154	561.39	587.91
K-Means + LR	0.8117	0.7255	0.7117	507.01	546.32
Linear SVM	0.8182	0.75	0.6923	—	—
Radial SVM	0.8636	0.8163	0.7692	—	—
KNN (K=5)	0.8701	0.7963	0.827	—	—

KNN

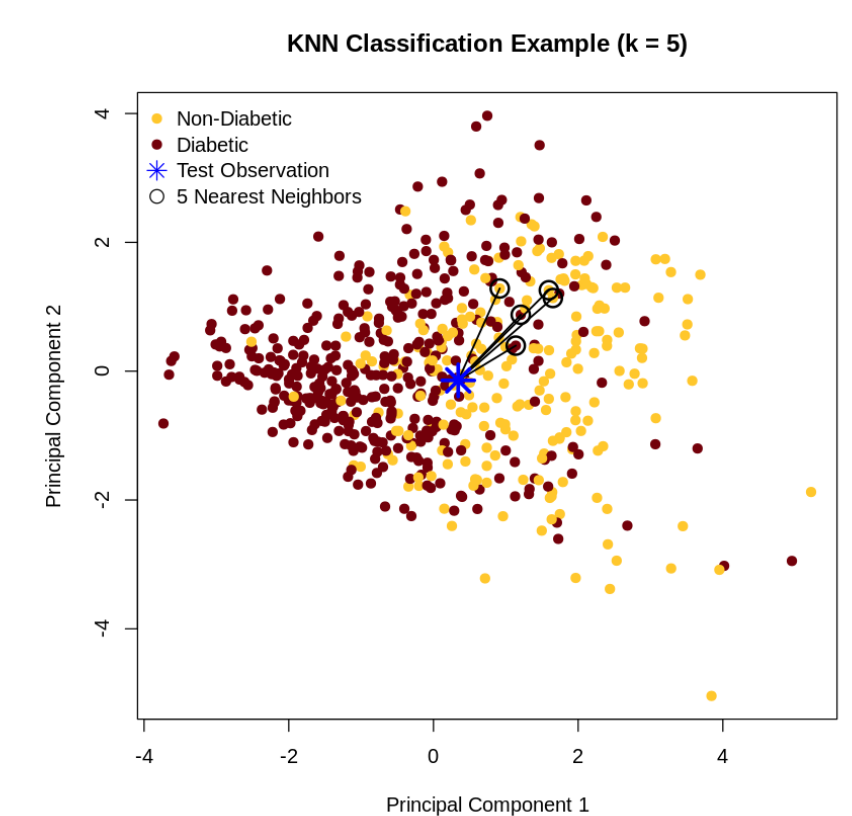
K-Nearest Neighbor (KNN) was used to evaluate the effectiveness of our preprocessing methods by comparing classification performance using accuracy, precision, recall, and F1-score.

Metric	Model
Accuracy	0.8701
Precision	0.7963
Recall	0.827
F1 Score	0.8113

KNN assigns a class based on the majority label of the k nearest neighbors.

Increasing k reduces sensitivity to noise but may smooth class boundaries.

This example illustrates how $k = 5$ classifies a new observation using local neighborhood information.

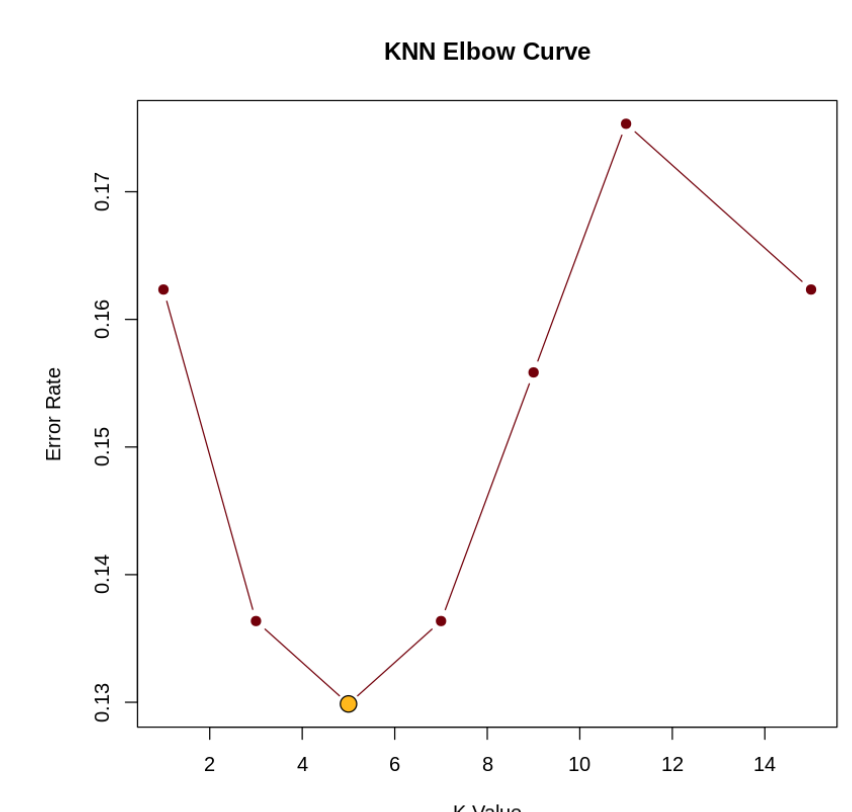


K Selection

Error rates were evaluated across multiple values of k .

The minimum error occurred at $k = 5$.

This value was selected for the final KNN classifier.



ACKNOWLEDGEMENTS

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